

CMR Specialist

Design Specs - Architecture - Capabilities - UI Guide (Short Manual, v2.0)

1. Design Principles

- **Store-first:** every input is persisted before any model call. No claim exists without a stored case, run, and step trail.
- **Model proposes, software enforces:** the model orders plans, extracts facts, drafts answers. The harness validates: allowlisted tools only, max 12 steps, patterned IDs regex-checked, approval gates, idempotency keys.
- **Graceful degradation:** model silence falls back to deterministic templates. Tasks keep moving; the fallback is flagged visibly (*plan_fallback*), never hidden.
- **Two tool systems:** internal tools (read/compute, unaudited) and the sealed tool module (soleDB writer, CHECK free / WRITE approval + audit).
- **Honest failure:** no evidence means the bounded fallback ('couldn't find supporting evidence'), never an invented answer.

2. Architecture

Web API (port 8080) + background worker + SQLite/Postgres store + operator console + tool module over a mock CMR DB. Inference via OpenRouter (configurable). Config never auto-loads *.env*: web needs `--env-file .env`, worker needs env-exported variables.

Layer	Role	Key files
Intake	Guard, intent (model-first + veto), extraction (model + regex floor)	guard.py, intent.py, extractor.py
Planning	Model orders steps from allowlist; harness enforces; plan stored at creation	planner.py, service.queue_task
Execution	Worker honors stored plan; policy gate; idempotent steps	worker.py, service.execute_task
Tool module	CHECK/WRITE formulations; only DB writer; append-only audit	tools/module.py, mockdb.py
Knowledge	Upload, chunk, publish; only <i>active</i> versions searchable	knowledge APIs

3. Capabilities

Task families

- **payment_reconciliation** - needs RRR (12 digits) + date + account. Verifies receipt against ledger.
- **change_of_ownership** - needs plate + buyer + seller names. Names come from the model; plate is regex-validated.
- **general_support** - knowledge-grounded answers with cited spans.
- **Knowledge Q&A** - answers cite published doc spans; empty base returns the bounded fallback.

Task lifecycle

queued (all facts present) - *waiting_for_input* (missing facts + prompts; plan already stored) - *waiting_approval* (sensitive/WRITE steps) - *running* - *completed* / *failed*. Follow-up messages merge new facts and resume waiting tasks automatically.

Visibility flags

- **plan_proposed_by:** *model* (reasoned) or *template* (fallback cover).
- **plan_fallback:** **true** means the template covered for a silent model - work continues, reasoning didn't happen. Clustered fallbacks signal quota/throttling, not code faults.

4. UI Guide (<http://localhost:8080>)

Card	What to do	What you get back
1 - New case	Paste a question or task, Submit	intent, task_id, status, missing facts + prompts, plan flags
2 - Follow-up	Case ID carries over; paste missing fact, Attach	merged facts; waiting tasks resume (<i>resumed=1</i>)
3 - Queue task	Pick type + approval level, Queue	direct task (use approval: always to see the gate)
4 - Knowledge	Upload file, then Publish the version id	chunk count; only published versions are searchable
Case state	Refresh case / Refresh task	fields, plan + <i>plan: model/template (fallback)</i> badge
Approval	Approve / Reject with reason	waiting_approval tasks proceed or stop

5. Two guided exercises

Exercise A - waiting then resume (ownership). Card 1:

Musa sold his car to Adaeze Okafor, please transfer the ownership

Expect *waiting_for_input* + missing plate. Then Card 2:

the plate number is ABC123XY

Expect resume, validation of the plate, then *completed*.

Exercise B - straight-through (payment). Card 1:

Please reconcile my Remita payment RRR 123456789012 made yesterday for account acct-X

Expect *queued* - *completed*, receipt verified paid (RRR ...9012), cited knowledge spans.

6. Troubleshooting

- Task stuck *queued*: worker down - restart it; check */health oldest_queued_job_age_s*.
- Always *waiting_for_input*: Refresh case shows the missing fact + prompt - answer via Card 2.
- Fallback badges everywhere: model throttled - check server log for *[planner]/[extractor]* lines; wait, retry.
- Port busy on restart: kill stale processes first (*pskill -f harness.api.main*), then launch web + worker.
- Knowledge answers empty: document uploaded but not *published* - publish the version id.